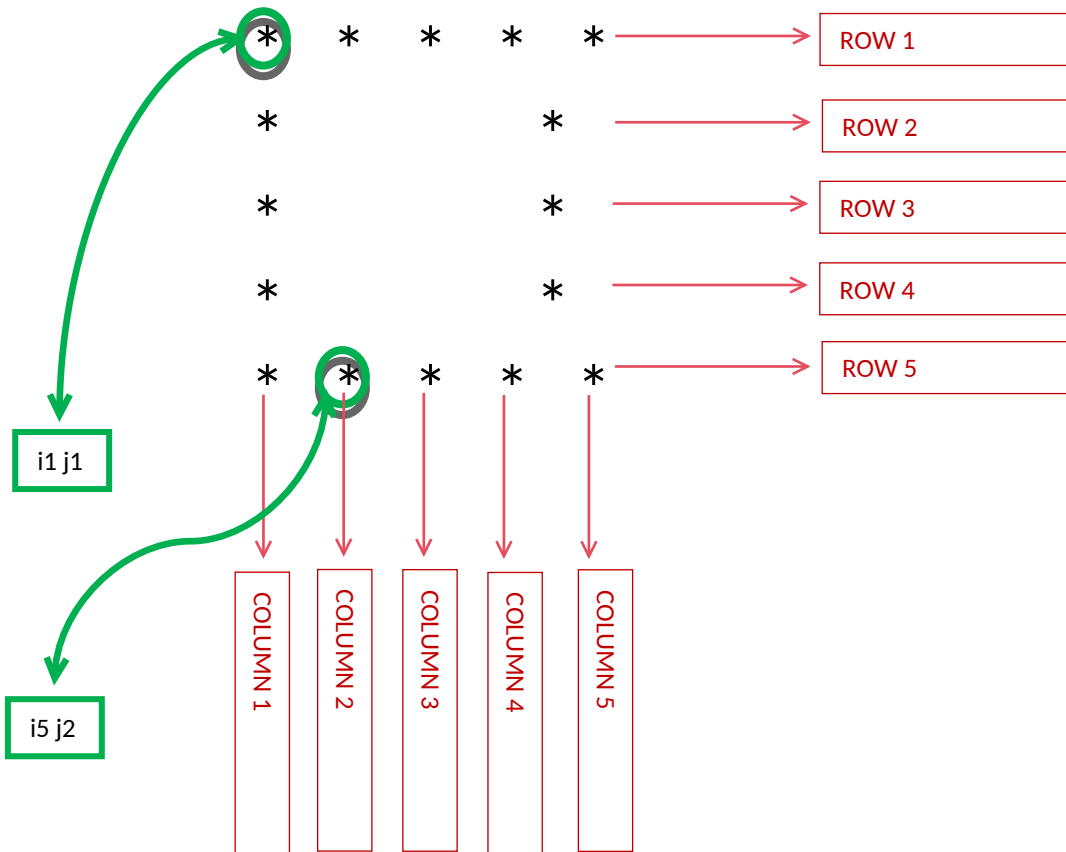


WEEK 01 MODULE
Programming Practice

TOPIC : 1 'Star Pattern'

2. HOLLOW SQUARE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms a square where the number of rows is equal to the number of columns.
- Unlike a filled square, only the boundary positions contain stars (*), while the inner positions remain empty.
- Therefore, the pattern consists of both printed and skipped positions.

ii. Execution Flow:

Step 1: Observe the Grid Concept:

- Pattern 2: Hollow Square

For Output (n = 5) figure obtained :-

```

*      *      *      *      *
*
*
*
*
*      *      *      *      *

```

- This is still a square.

- Rows = n
- Columns = n

- size of matrix = $n \times n$.

- Let's mark where stars appear :-

Stars exist only on the boundary.

The interior is empty.

For n= 5

✦ Boundary Analysis

Let's identify the four boundaries.

Top Boundary

```
*****
```

This is the first row.

```
i == 1
```

Bottom Boundary

```
*****
```

This is the last row.

```
i == n
```

Left Boundary

```

*
*
*
*
*

```

This is the first column.

```
j == 1
```

Right Boundary

*
*
*
*
*

This is the last column.

`j == n`

Most Important Observation

A star should be printed when the current cell belongs to any boundary given below i.e :-

Top Boundary

OR

Bottom Boundary

OR

Left Boundary

OR

Right Boundary

Translated into conditions:

`i == 1`

OR

`i == n`

OR

`j == 1`

OR

`j == n`

- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Step 3: For every row generated by the outer loop, traverse all columns of that row using the inner loop.

-The inner loop is responsible for generating columns.

Step 4: At each cell position, determine whether the cell lies on the boundary of the square.

Step 5: Otherwise, print a space to maintain the hollow structure.

Step 6: After all columns of the current row have been processed, move the cursor to the next line.

Step 6: Repeat the same process until all n rows have been completed.

2. Condition Analysis :-

Conditional statements are required because not every cell contains a star.

A star must be printed when:

- `i == 1`
- `i == n`
- `j == 1`
- `j == n`

These conditions represent the four boundaries of the square.

All remaining cells are interior cells and therefore contain spaces.

3. Program :-

```
package week1_module;
import java.util.Scanner;
public class HollowStarSquare {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of the square: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= n; j++) {

                if(i == 1 || i == n || j == 1 || j == n) {
                    System.out.print("* ");
                }
                else {
                    System.out.print(" ");
                }

            }

            System.out.println();
        }

        sc.close();
    }
}
```

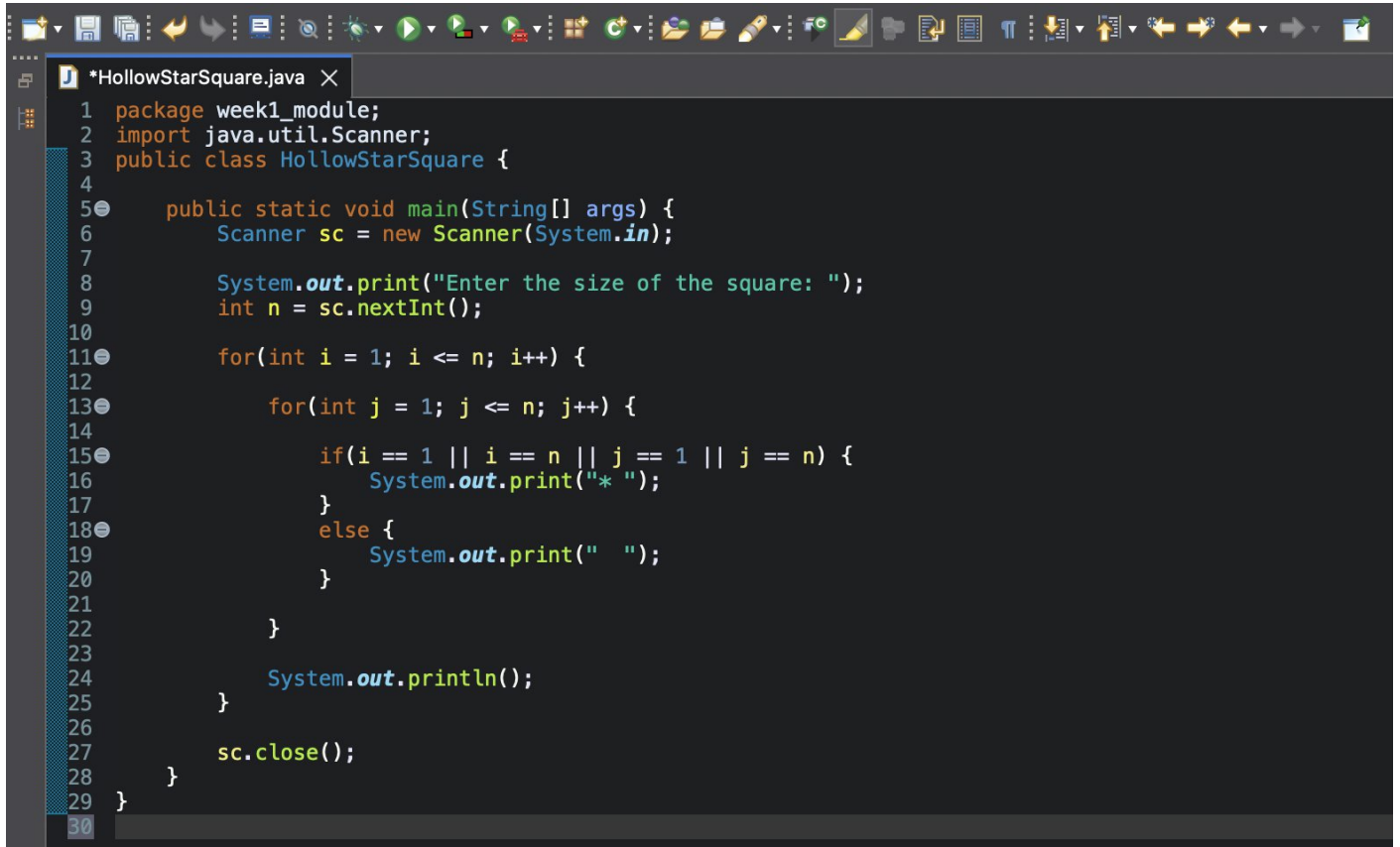
4. Program Output :-

Enter the size of the square: 5

```
* * * * *  
*      *  
*      *  
*      *  
* * * * *
```

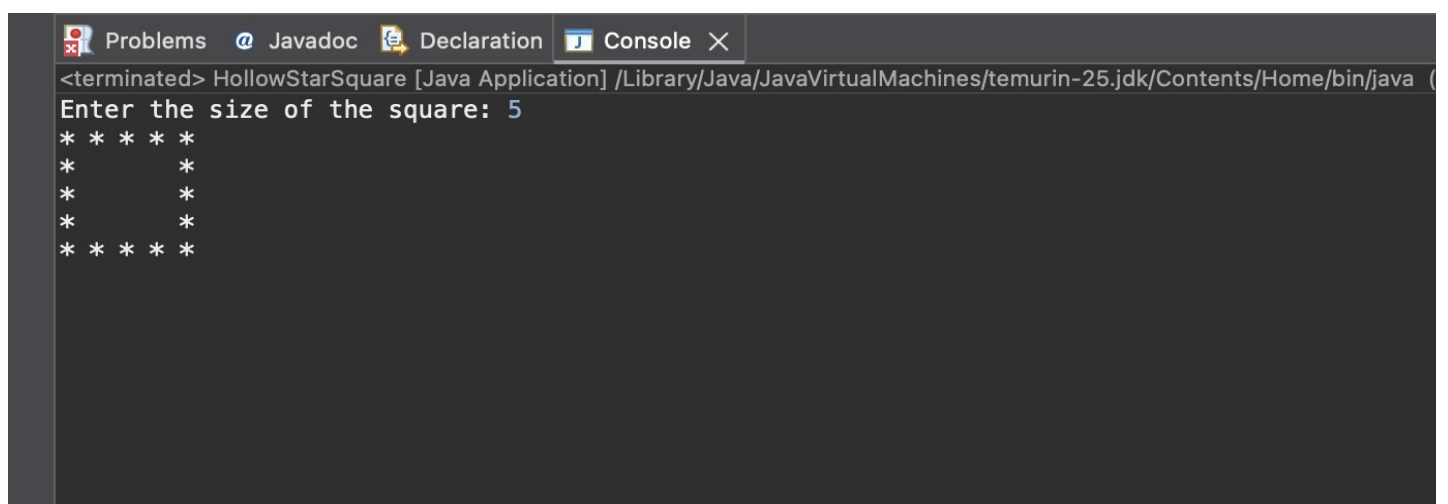
5. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class HollowStarSquare {
4
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the size of the square: ");
9         int n = sc.nextInt();
10
11         for(int i = 1; i <= n; i++) {
12             for(int j = 1; j <= n; j++) {
13
14                 if(i == 1 || i == n || j == 1 || j == n) {
15                     System.out.print("* ");
16                 }
17                 else {
18                     System.out.print(" ");
19                 }
20             }
21             System.out.println();
22         }
23         sc.close();
24     }
25 }
26
27
28
29
30
```

ii. Program's Output :-



```
<terminated> HollowStarSquare [Java Application] /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/java (
Enter the size of the square: 5
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```